

Pedro Castro

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PROFILE

Informatics and Computing Engineering student focused on backend engineering and infrastructure. Experienced with Java/Kotlin, Spring Boot and Docker. Passionate about building reliable software, exploring new technologies, and working on challenging problems. Open-source supporter.

EDUCATION

Faculty of Engineering of the University of Porto <u>Master in Software Engineering</u>	Porto, Portugal Sep 2026–Present
Relevant Coursework: Software Architecture, Requirements Engineering, Software Testing, AI in Software Engineering, Secure Software Engineering	
<u>Bachelor in Informatics and Computing Engineering</u>	Sep 2022–Jul 2026
Relevant Coursework: Algorithms and Data Structures, Databases, Web Development, Operating Systems, Software Engineering, Distributed Systems, Computer Security, Compilers, Discrete Mathematics, Artificial Intelligence.	

EXPERIENCE

NIAEFEUP <u>Co-Head of Projects</u>	Porto, Portugal May 2026–Present
- Coordinate and support multiple software projects across the student organization. - Work with project managers on planning, communication, and delivery across all teams. - Co-Head of Informatics for <u>Semana de Informática</u>, the organization's flagship annual tech event.	
<u>Project Manager</u>	Aug 2025–May 2026
- Led development of the association's website, coordinating a team of 15+ people across backend and frontend. - Maintained/extended the association's backend (Kotlin, Spring Boot, PostgreSQL) and frontend (Svelte, Tailwind CSS).	
<u>Member</u>	Oct 2024–Present
- Contributor to <u>uni</u>, a Flutter app helping students organize academic life and schedules. - Contributed to infrastructure for the organization's Kubernetes cluster (<u>niployments</u>). - Member of <u>SINF's</u> Competitions team, designing CTF challenges and programming contest problems.	
Faculty of Engineering of the University of Porto <u>Teaching Assistant</u>	Porto, Portugal Feb 2026–Jun 2026
Teaching assistant for the Computer Laboratory course (C device driver implementation in MINIX), supporting 40+ students with system programming and debugging.	
Kevel <u>Engineering Intern</u>	Porto, Portugal Jul 2025
Designed and built Scala tooling for CI pipelines to detect and report non-deterministic tests in large backend codebases, improving reliability and preventing flaky builds from reaching production.	

PROJECTS

NixOS Configuration <i>Nix, Linux, Infrastructure as Code</i>	Present
- Built a declarative infrastructure-as-code system for managing multiple NixOS machines with a modular configuration. - Deployed a home server running 15+ self-hosted services including media streaming, photo management, and file storage. - Configured reverse proxy with automated SSL certificate management via Cloudflare DNS for remote access.	
Personal Website/Portfolio <i>Astro, Tailwind CSS</i>	Present
Personal portfolio built with Astro and Tailwind CSS to showcase projects and host my resume.	
DistributedChat <i>Java, TLS, PostgreSQL, Distributed Systems</i>	May 2026
- Developed a TLS-secured distributed chat system with Java virtual threads, PostgreSQL persistence, and session tokens. - Designed for fault tolerance with automatic retry, state preservation across disconnects, and integrated a local LLM (Ollama) for AI-powered responses in chat rooms.	
OpenQuest <i>Laravel, PostgreSQL, REST APIs, OpenAPI</i>	Dec 2025
- Created a collaborative Q&A platform with a REST API documented in OpenAPI and role-based access control. - Designed a PostgreSQL schema for questions, answers, voting, notifications, badges, and content moderation.	

TECHNICAL SKILLS

Languages: C, C++, Java, Kotlin, Scala, Python, JavaScript, SQL, Dart, PHP, Nix
Backend: Spring Boot, NestJS, Laravel, REST APIs, OpenAPI
Databases: PostgreSQL, MongoDB, SQLite, MySQL, Database Design, Relational Algebra
DevOps & Tools: Git, Docker, CI/CD, Linux, Nix, Caddy
Frontend: Svelte, Astro, Tailwind CSS
Software Engineering: OOP, Design Patterns, Software Architecture, Unit Testing, Refactoring, Debugging

LANGUAGES

Portuguese: Native | **English:** Highly Proficient